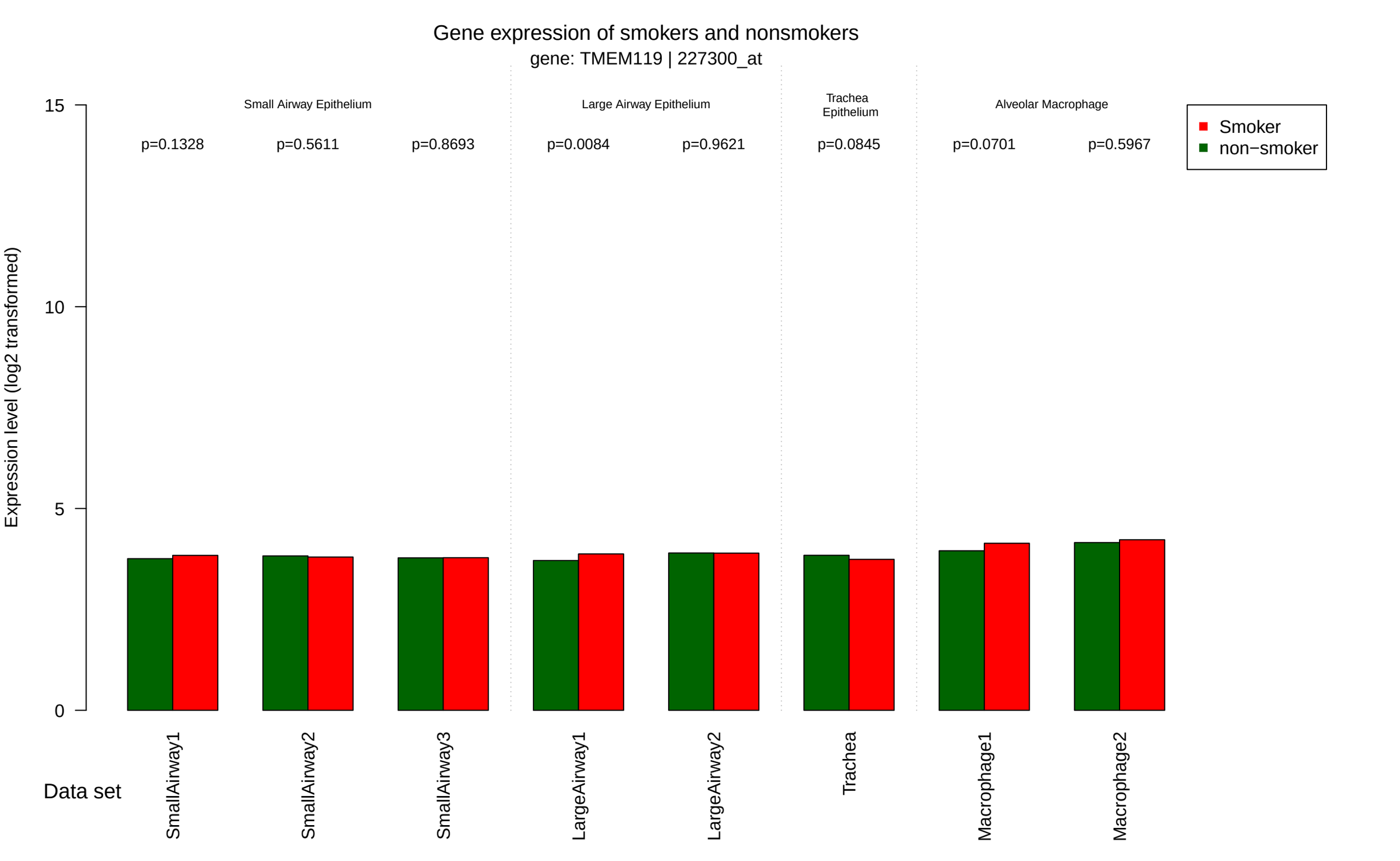
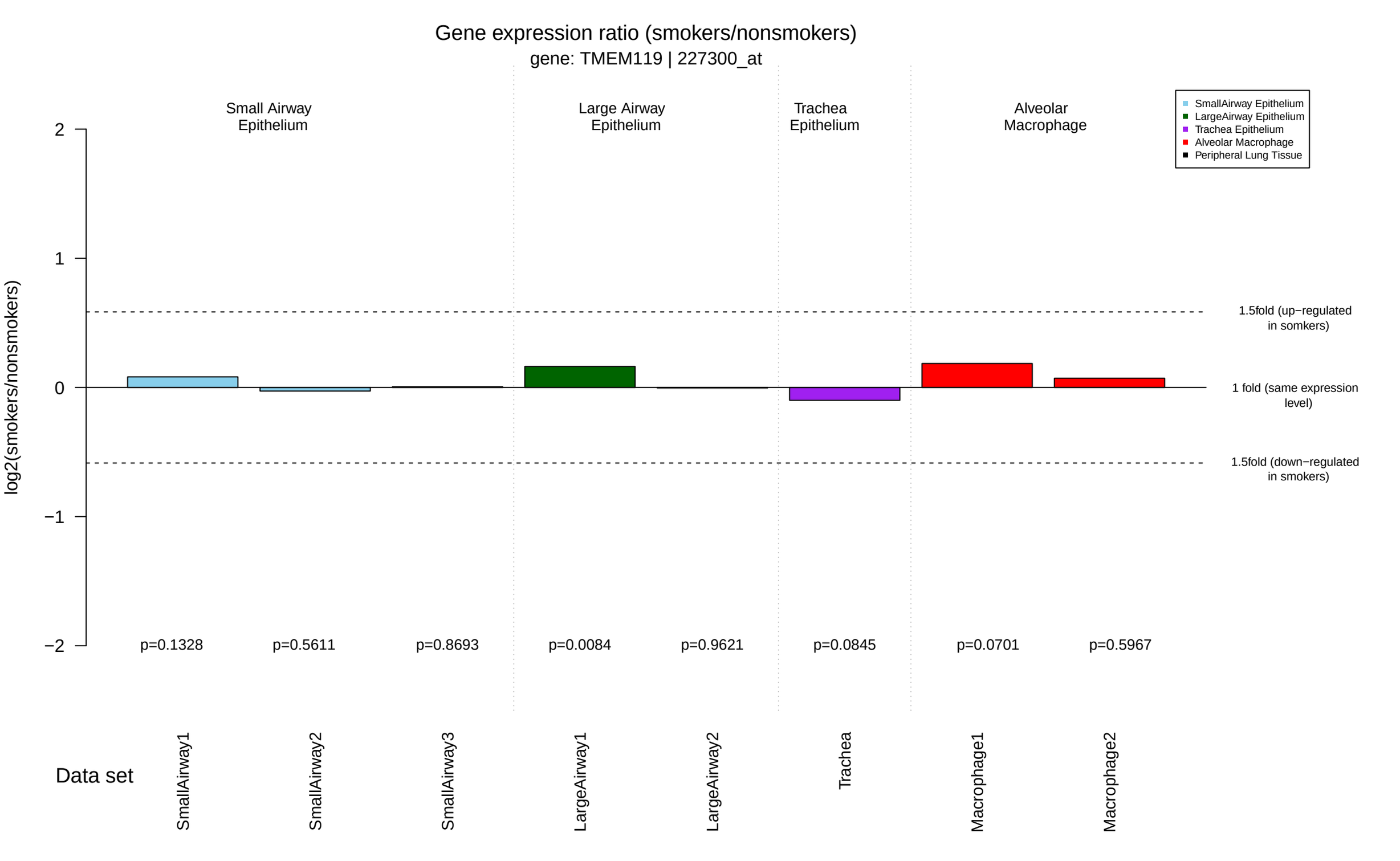
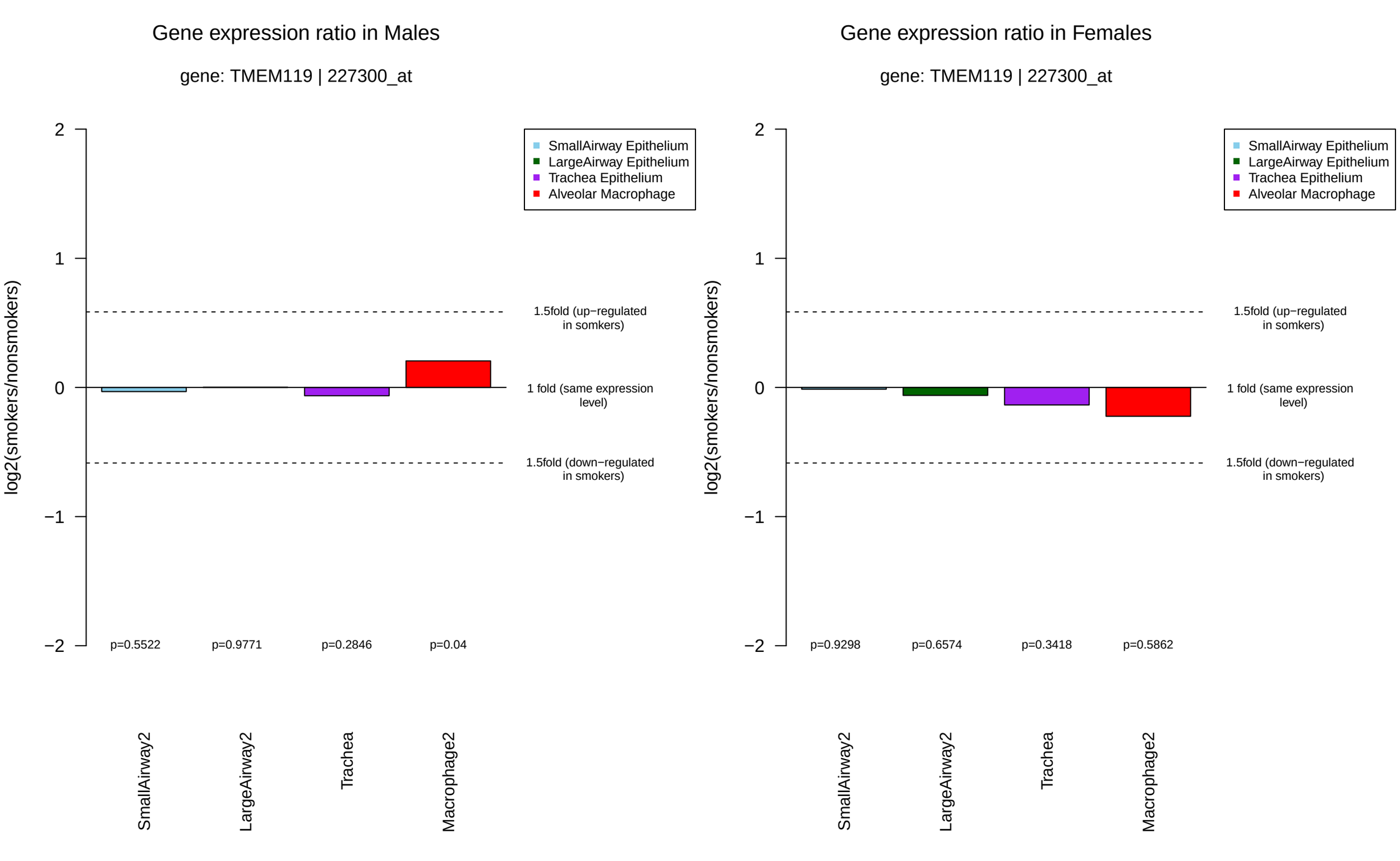
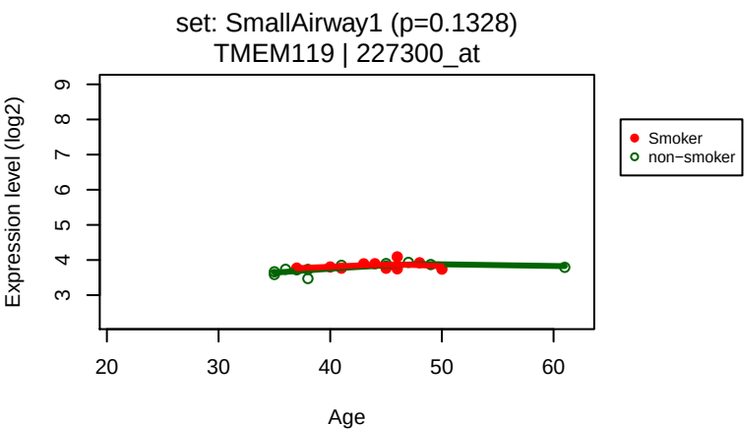
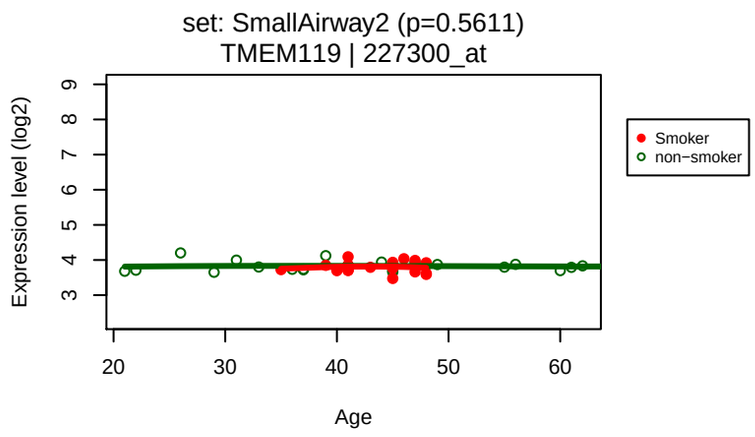
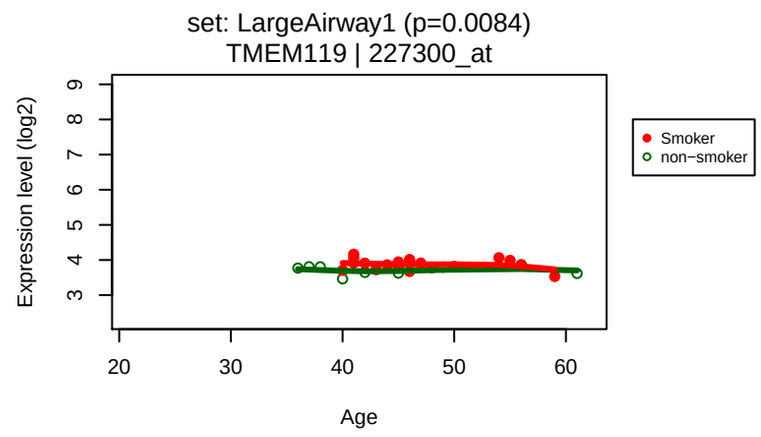

Gene expression of smokers and nonsmokers
gene: TMEM119 | 227300_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1328
p=0.5611
p=0.8693
p=0.0084
p=0.9621
p=0.0845
p=0.0701
p=0.5967
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

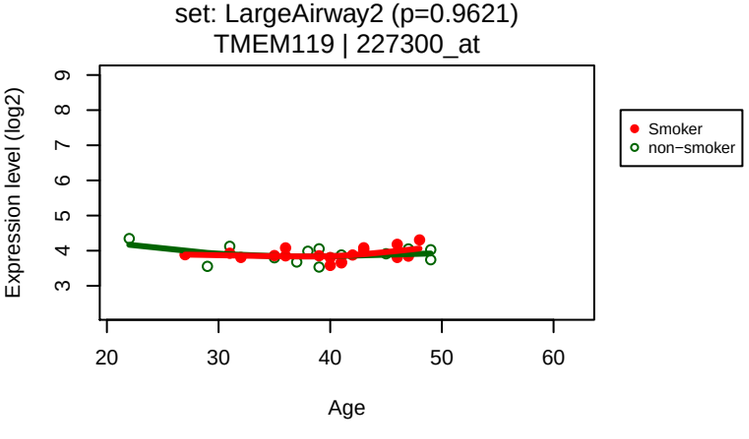

Gene expression ratio (smokers/nonsmokers)
gene: TMEM119 | 227300_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1328
p=0.5611
p=0.8693
p=0.0084
p=0.9621
p=0.0845
p=0.0701
p=0.5967
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

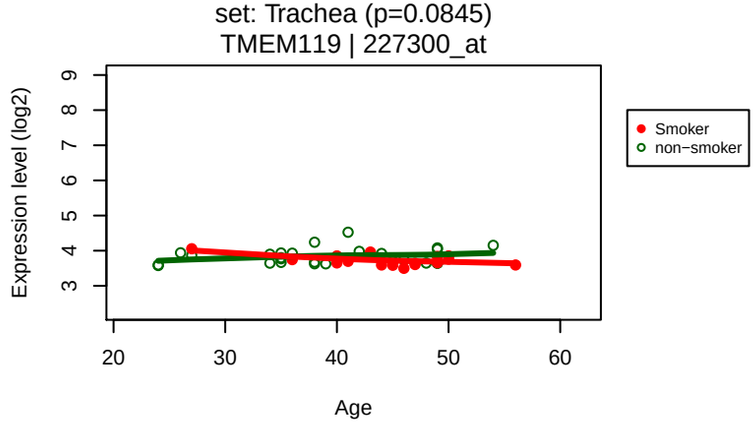

Gene expression ratio in Males
gene: TMEM119 | 227300_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5522
p=0.9771
p=0.2846
p=0.04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM119 | 227300_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9298
p=0.6574
p=0.3418
p=0.5862
SmallAirway2
LargeAirway2
Trachea
Macrophage2

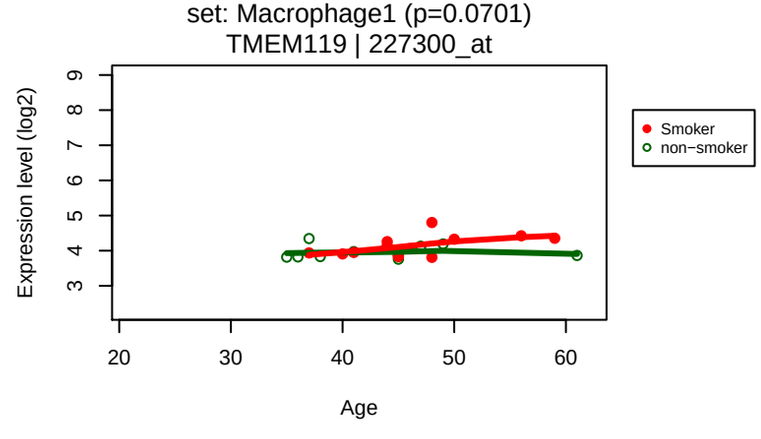

set: SmallAirway1 (p=0.1328)
TMEM119 | 227300_at
Expression level (log2)
Age
Smoker
non−smoker

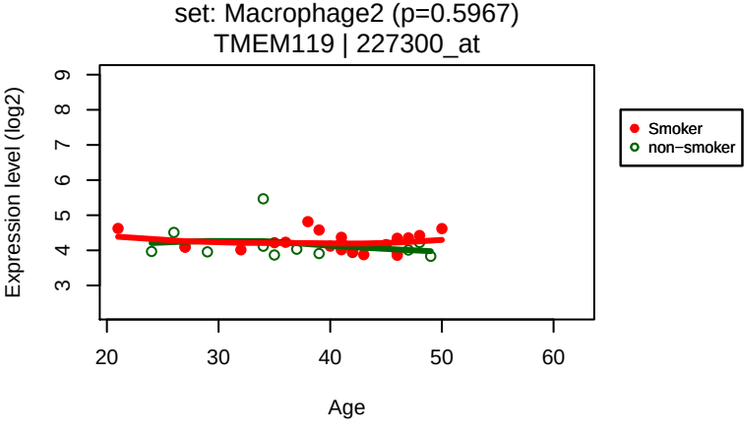

set: SmallAirway2 (p=0.5611)
TMEM119 | 227300_at
Expression level (log2)
Age
Smoker
non−smoker

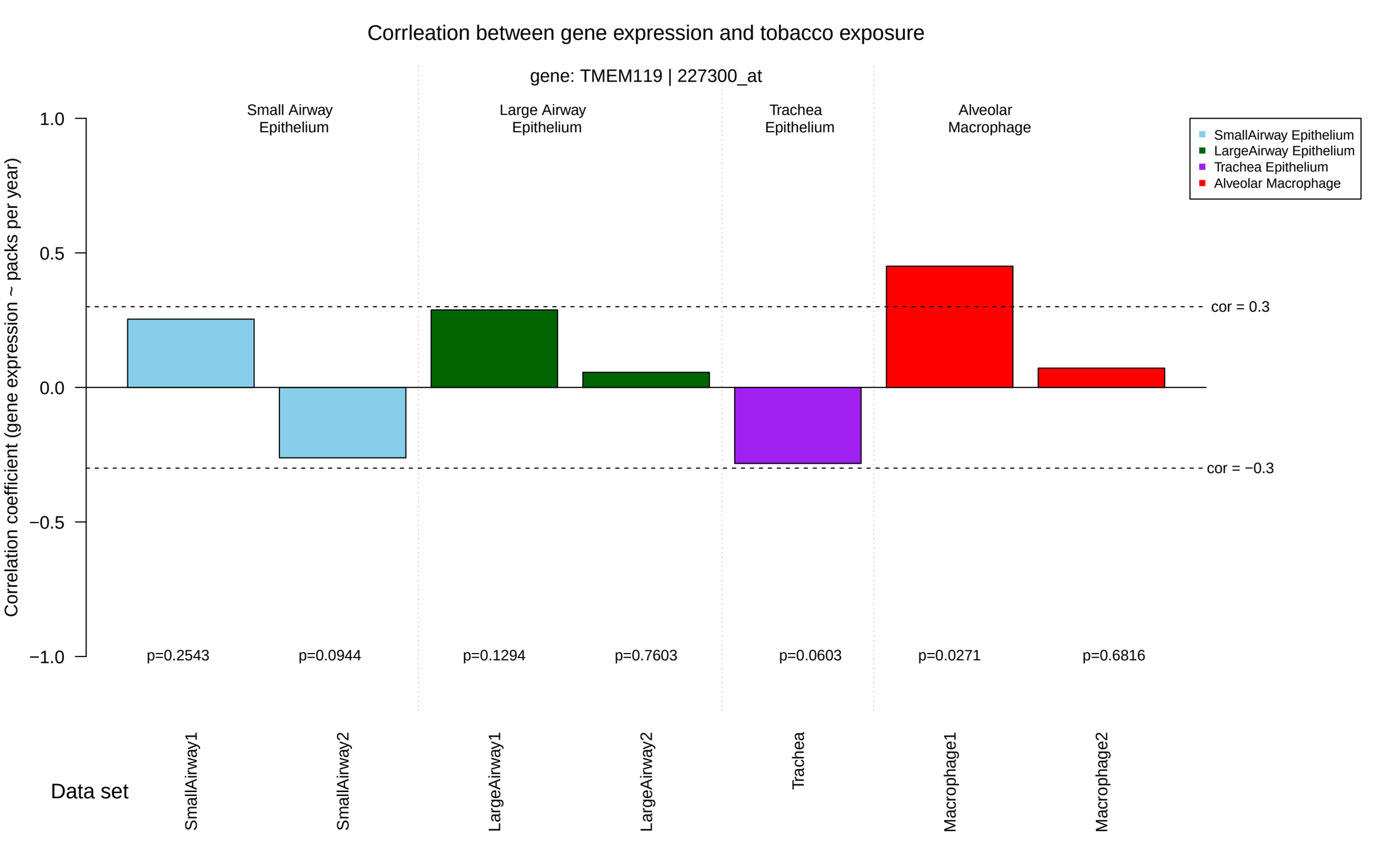

set: LargeAirway1 (p=0.0084)
TMEM119 | 227300_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9621)
TMEM119 | 227300_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0845)
TMEM119 | 227300_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0701)
TMEM119 | 227300_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5967)
TMEM119 | 227300_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM119 | 227300_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2543
p=0.0944
p=0.1294
p=0.7603
p=0.0603
p=0.0271
p=0.6816
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2